INTEGRATED SPEECH PROCESSING SYSTEM

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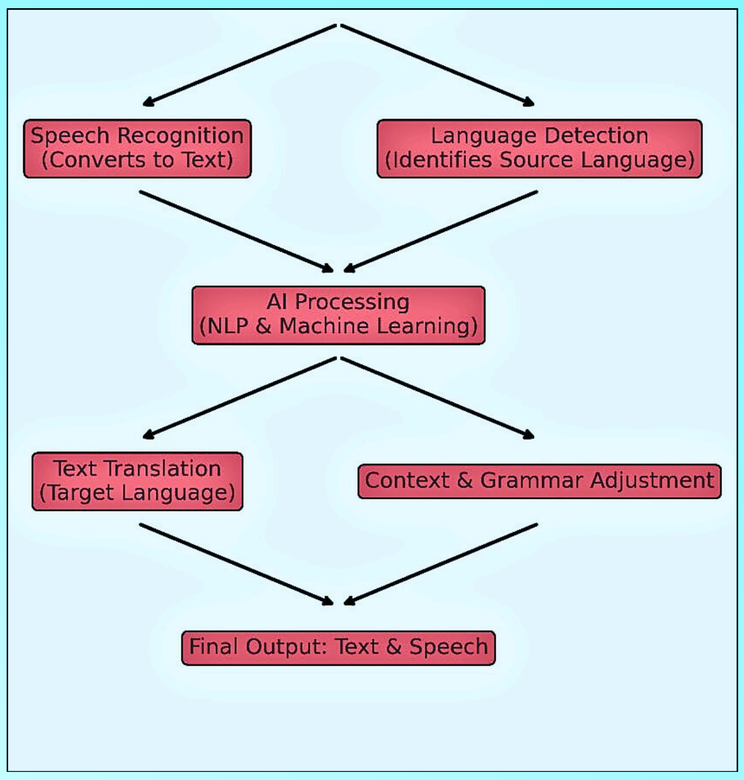
FLOWCHART



01. INTRODUCTION

Speech-to-text and text translation are essential in modern NLP systems, powering applications like voice assistants, real-time transcription, and multilingual tools. This project focuses on building a modular system that sequentially converts raw audio to text and translates it into another language. The modular design enables independent enhancement of each component.

02. METHODOLOGY

🎙 2.1 Speech-to-Text (STT) Model

Purpose: Converts raw audio into text.

- Input: Raw audio waveforms.
  - Preprocessing: Spectrogram Conversion: Uses STFT to convert audio into spectrograms for better feature extraction.
  - Normalization: Ensures consistent input by adjusting to zero mean and unit variance.
  - Padding: Standardizes input length.
  - Model Architecture:Conv2D Layers: Extract local features like phonemes.
  - Bidirectional LSTM Layers: Capture temporal dependencies in both directions.
  - Dense Layer: Outputs character probabilities.
- Loss Function: CTC (Connectionist Temporal Classification) for handling non-aligned input-output sequences.
  - Training: Dataset: LJSpeech 1.1 (13,100 audio clips).
  - Batch Size: 8 | Learning Rate: 0.0005 | Epochs: 1000

🌐 2.2 Text Translation Model

Purpose: Translates text generated by the STT model.

- Input: Transcribed text (e.g., "Hello, how are you?").
  - Preprocessing: Tokenization: Uses Byte Pair Encoding (BPE) for handling rare words.
  - Padding: Standardizes input sequence length.
  - Model Architecture: Transformer: Utilizes self-attention mechanisms.
  - Encoder: Processes input text with multi-head self-attention.
  - Decoder: Generates translated text, focusing on context and previous tokens.
  - Positional Encoding: Adds token order information.
- Loss Function: Cross-entropy loss for accurate translation.
  - Training: Dataset: WMT14 (English, Hindi, French).
  - Batch Size: 64 | Learning Rate: 0.0001 | Epochs: 50

03. RESULTS/FINDINGS

3.1 Speech to Text Model

CER : 8.5% → This means that, on average, 8.5% of the characters in the predicted text are incorrect.

WER : 12.3% → This means that, on average, 12.3% of the words in the predicted text are incorrect.

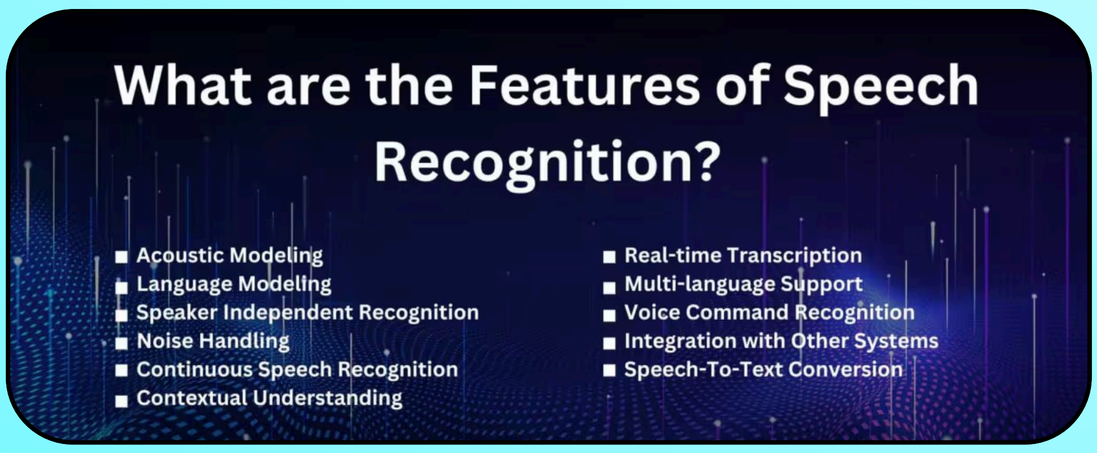
The model performs well on clean audio but struggles with noisy or low quality recordings.

3.2 Text Translation Model

BLEU Score : 28.4

This score indicates that the translations are reasonably accurate but can be improved further.

The model performs well on common phrases but struggles with rare words and complex sentence structures.



04. CONCLUSION

THIS PROJECT DEMONSTRATES THE FEASIBILITY OF BUILDING AN END TO END SYSTEM FOR SPEECH TO TEXT AND TEXT TRANSLATION. THE MODULAR DESIGN ALLOWS FOR INDEPENDENT IMPROVEMENT OF EACH COMPONENT. FUTURE WORK INCLUDES INTEGRATING THE TWO MODELS INTO A SINGLE PIPELINE AND OPTIMIZING FOR REAL TIME PERFORMANCE.

05. REFERENCES

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